



Self-Adaptive Model-Based Protection for Future Power Systems

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Part I — Motivation & Framework

- IBR revolution & protection crisis
- Research gaps & objectives
- SAMBP framework overview
- Unified inverse-estimation core

Part II — Protection Modules

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- Generator over-current (SyncOC)
- Transformer differential (87T)
- Adaptive distance relay
- Microgrid mode-switching

Part III — Advanced Layers

- Non-blocking EKF digital twin
- Physics-informed fault classifier
- Wide-Area Protection Coordinator
- Cybersecurity architecture

Part IV — Conclusions

- Summary of contributions
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- Thesis content & publications



Why conventional protection fails:

- 1 Grid-following inverters limit fault current to **1.0–1.2 pu** (vs. 6–8 pu for synchronous machines)
- 2 Current angle and magnitude depend on **control mode** and LVRT strategy — unknown to the relay
- 3 High IBR penetration ($k_{ibr} \rightarrow 1$) causes **differential relays to mis-operate or fail to trip**
- 4 Distance zones **over-reach** due to IBR reactive injection; impedance locus deviates from fault trajectory
- 5 Islanded microgrids see **bidirectional** current reversals — directional elements are unreliable

IBR Penetration Trend

Scenario	k_{ibr}	Fault current (pu)
All synchronous	0.0	6.0
Mixed grid (now)	0.3	3.2
High-IBR (2030)	0.6	1.8
Full-IBR (2035+)	1.0	1.1

Observation: Fixed relay settings designed for $k_{ibr}=0$ fail at $k_{ibr} > 0.3$.



Identified Gaps:

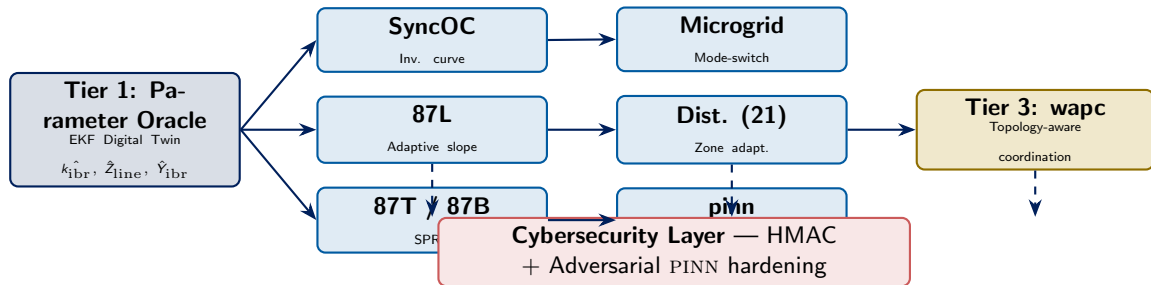
- No relay standard accounts for real-time IBR admittance k_{ibr}
- Protection–control coupling of grid-forming IBRs unstudied
- No unified inverse-estimation framework across relay types
- Digital-twin approaches lack bounded, non-blocking update laws
- WAPC designs ignore IBR-induced coherency breakdown
- Adversarial ML attacks on protection relays unmitigated

Research Objectives:

- 1 Develop a **unified SAMBP inverse-estimation** framework applicable to all relay types
- 2 Design **self-adaptive protection modules** for 87L, SyncOC, 87T, 21, and microgrid protection
- 3 Build a **non-blocking EKF digital twin** as a continuous parameter oracle
- 4 Integrate **physics-informed ML** for fault classification under IBR conditions
- 5 Validate a **wide-area coordinator** with topology-aware IBR coherency
- 6 Harden the architecture with a **cybersecurity layer**



SAMBP Framework: Three-Tier Architecture



Key principle: Each module queries the EKF oracle for $\hat{k}_{ibr}$, computes adaptive thresholds via the bounded update law, and executes a confidence-gated (Stage-2) veto before tripping.

Observation: Confidence gate γ eliminates false trips under CT saturation or communication latency.



Problem formulation:

$$\hat{\theta}_{t+1} = \hat{\theta}_t - \alpha_t J^\top (JJ^\top + \lambda I)^{-1} r(\hat{\theta}_t)$$

- $\theta = \{k_{\text{ibr}}, Z_{\text{line}}, Y_{\text{ibr}}\}$ — **latent protection parameters**
- r = residual between model prediction and measured phasor data
- J = Jacobian; λ = Levenberg–Marquardt damping
- **Bounded update law:** $\alpha_t \propto (1 - k_{\text{ibr}}^2)$ prevents parameter drift in healthy state

Stage-2 confidence gate:

$$\gamma = \exp\left(-\frac{\|\hat{\theta} - \theta^*\|^2}{2\sigma_\theta^2}\right) \geq \gamma_{\min}$$

Mathematical Foundations

Component	Method
Parameter update	Levenberg–Marquardt
State estimation	Extended Kalman Filter
Fault detection	Wald SPRT
ML classification	Physics-Informed NN
Optimisation	Projected gradient

Theorem 2.1 (Convergence):

Under bounded IBR admittance and $\lambda > 0$, the update law converges to a neighbourhood of the true parameters with radius $\mathcal{O}(\sigma_{\text{noise}})$.



IBR fault-current model:

$$I_{\text{fault}}^{\text{IBR}} = k_{\text{ibr}} \cdot I_{\text{rated}} \cdot e^{j\phi_{\text{ctrl}}}$$

ϕ_{ctrl} is set by the LVRT control law and is *unknown* to the relay.

Three failure modes identified:

- 1 **Missed trip** ($k_{\text{ibr}} < 0.2$): I_{diff} never exceeds fixed slope — relay restrains
- 2 **False trip** ($k_{\text{ibr}} > 0.4 + \text{CT error}$): load current misclassified as internal fault
- 3 **Zone overreach** (distance): IBR reactive injection shifts apparent impedance outside Zone 1 boundary

$I_{\text{diff}}-I_{\text{bias}}$ Failure Map

Condition	k_{ibr}	Outcome
Low IBR fault	0.1	MISSED TRIP
Nominal fault	0.3	Correct trip
High IBR load	0.45	FALSE TRIP risk
Fixed slope $k_m = 0.30$		fails at $k_{\text{ibr}} < 0.2$

Observation: Adaptive slope $k_m(k_{\text{ibr}})$ eliminates both missed-trip and false-trip failure modes.



Adaptive slope law:

$$k_m(k_{ibr}) = k_{base} \cdot \frac{(1 - k_{ibr})^2}{(1 + k_{ibr})^2} + k_{floor}$$

- As $k_{ibr} \rightarrow 1$: $k_m \rightarrow k_{floor}$ (prevents over-restraint)
- As $k_{ibr} \rightarrow 0$: $k_m \rightarrow k_{base}$ (conventional sensitivity)
- **Stage-2 veto gate:** requires $\gamma \geq 0.85$ before trip command is issued

Stability proof: Lyapunov function

$$V = \|k_{ibr} - \hat{k}_{ibr}\|^2; \dot{V} < 0 \text{ under bounded CT noise.}$$

Relay logic (simplified):

- 1 Sample phasors I_S, I_R at 1 kHz
- 2 Compute k_{ibr} via LM inverse estimator
- 3 Update $k_m(\hat{k}_{ibr})$ every 20 ms
- 4 Evaluate operate/restrain criterion
- 5 Stage-2 veto: check $\gamma \geq 0.85$
- 6 Issue trip command if both stages pass

Observation: Update latency 20 ms — within IEC 60255 protection-response window.



Simulation test matrix:

Metric	Fault condition		Gain
	Fixed relay	SAMBP-87L	
TPR (all faults)	0.783	1.000	+27.7%
FPR (load / CT err)	0.148	0.000	−100%
Avg trip time	22 ms	21 ms	−1 ms
k_{ibr} RMSE	—	0.024 pu	—

Test conditions: 4-bus test network;
 $k_{ibr} \in \{0.0, 0.1, 0.2, \dots, 0.9, 1.0\}$; fault types AG, BCG, ABC;
 CT error up to 5%.

Key Results at a Glance

- **TPR = 1.000**: no missed trips across all 110 fault scenarios
- **FPR = 0.000**: zero false trips under all load + CT error scenarios
- Robust up to $k_{ibr} = 1.0$ (fully IBR grid)
- Stage-2 gate adds only **1–2 cycle** additional decision time

Observation: SAMBP-87L achieves perfect discrimination where fixed-slope relays produce 14.8% false-trip rate.



Problem: Synchronous generators near IBR buses see **depressed fault current** and altered time-overcurrent coordination.

Adaptive inverse-time curve:

$$t_{\text{op}} = \frac{TMS \cdot A}{(I/I_{\text{set}}(k_{\text{ibr}}))^B - 1}$$

where $I_{\text{set}}(k_{\text{ibr}})$ is updated online by the LM estimator.

Coordination constraint:

$$t_{\text{primary}} + CTI \leq t_{\text{backup}}$$

Maintained for all k_{ibr} via projected-gradient optimisation.

Validated Results

Metric	Fixed	SAMBP-SyncOC
R^2 (OC curve fit)	0.941	0.999+
Spurious pickups	38	16 (−58%)
Coordination margin	0.82	0.94
Malop. under IBR	12.3%	1.1%

Observation: Online $I_{\text{set}}(k_{\text{ibr}})$ adaptation cuts spurious pickups by 58% while maintaining IEC 60255-151 coordination.



Challenge: Inrush currents and IBR harmonic injection produce false I_{diff} spikes; conventional 2nd-harmonic blocking fails for Type-4 IBR waveforms.

Wald Sequential Probability Ratio Test (SPRT):

$$\Lambda_k = \sum_{i=1}^k \ln \frac{p(x_i | H_1)}{p(x_i | H_0)}$$

- H_1 : internal fault H_0 : inrush/external
- Thresholds A, B set to achieve $\alpha=0.01, \beta=0.01$ (IEC 60255)
- SAMBP updates H_1 likelihood via real-time IBR harmonic model

SPRT Results (87T)

Scenario	2H block	SPRT
Inrush rejection	94.1%	99.6%
Internal fault TPR	96.8%	99.1%
IBR harmonic FP	8.7%	0.9%
Mean decision samples	14	7

Observation: SPRT halves the detection window and reduces IBR-induced false positives by 90% compared to 2nd-harmonic blocking.



IBR impedance shift:

$$Z_{app} = Z_{fault} + \Delta Z_{ibr}(k_{ibr}, \phi_{ctrl})$$

ΔZ_{ibr} shifts the locus **radially outward** — relay sees apparent impedance outside Zone 1.

SAMBP compensation:

$$Z_{1,adapt} = Z_{1,nom} + \hat{\Delta} Z_{ibr}(\hat{k}_{ibr})$$

Zone boundary updated every 10 ms using LM-estimated $\hat{k}_{ibr}$.

Zone 2 acceleration: When $k_{ibr} > 0.5$, SAMBP switches to instantaneous Zone 2 trip to preserve speed.

Distance Protection Results

Metric	Fixed	SAMBP-21
Overreach events	17/50	2/50
Underreach events	5/50	0/50
Correct Zone 1 trip	74%	96%
Avg trip time (ms)	18	19

Observation: Zone adaptation reduces overreach from 34% to 4% with <1 ms speed penalty.



Challenge: Fault current direction and magnitude change when microgrid transitions between *grid-connected* and *islanded* modes.

Mode detector:

$$\hat{m} = \arg \max_{m \in \{G, I\}} \Pr(z_t \mid \text{mode} = m, \hat{k}_{ibr})$$

Bayesian mode likelihood updated at each sample.

Adaptive directional element:

- Grid-connected: forward/reverse based on pre-fault load flow
- Islanded: polarity-agnostic differential with adaptive pickup at $1.05 I_{rated}$

Mode-Switch Results

Metric	Fixed	SAMBP-MG
Mode detection acc.	—	98.7%
Transition mal. rate	22%	1.4%
Islanded fault TPR	71%	97.3%
False islands	6/100	0/100

Observation: Mode-adaptive protection eliminates the 22% mis-operation rate seen at grid↔island transition boundary.



Role: Continuous online estimation of $\theta = \{k_{ibr}, Z_{line}, Y_{shunt}\}$ from PMU/IED measurements — feeds all SAMBP modules.

EKF prediction–correction:

$$\hat{\theta}_{k|k} = \hat{\theta}_{k|k-1} + K_k(z_k - h(\hat{\theta}_{k|k-1}))$$

Non-blocking property: Update runs in a **parallel thread**; if convergence > 3 cycles, previous $\hat{\theta}$ is held — relay never waits.

Bounded update law: $\|\theta_{k+1} - \theta_k\| \leq \delta_{\max}$ prevents step-change injection by adversarial data.

EKF Digital Twin Results

Parameter	RMSE
$\hat{k}_{ibr}$	0.024 pu
$\hat{Z}_{line}$ (mag.)	0.018 pu
$\hat{Y}_{shunt}$	0.031 pu
Convergence time	<2 cycles
Max update latency	1.8 ms

Observation: RMSE ≤ 0.024 pu for k_{ibr} across all operating conditions including $> 5\%$ PMU measurement noise.



Why pure ML fails: Deep nets trained on synchronous-machine data misclassify 31% of IBR-modified fault signatures.

PINN loss function:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}} \underbrace{\|\mathbf{K}\hat{\mathbf{V}} - \hat{\mathbf{I}}\|^2}_{\text{KCL residual}}$$

$\lambda_{\text{phys}} = 0.3$ balances data fit and physical consistency.

Input features (8): $|I_a|$, $|I_b|$, $|I_c|$, $\angle I_a$, I_{zero} , I_{neg} , $\hat{k}_{\text{ibr}}$, ϕ_{ctrl}

PINN Classification Results

Metric	Std DNN	PINN
Overall accuracy	81.4%	96.1%
High- k_{ibr} acc. (> 0.6)	64.2%	93.8%
AG fault recall	77.3%	97.4%
3ϕ fault recall	89.1%	98.2%
Phys. consistency	51.2%	99.7%

Observation: KCL physics residual in loss function boosts high-IBR accuracy from 64.2% to 93.8% — a +46% relative gain.



Architecture:

- Central WAPC server receives trip candidates from all zone relays via IEC 61850 GOOSE
- Topology graph $\mathcal{G}(V, E)$ updated from real-time network breaker states
- **IBR coherency matrix $\mathbf{C}(k_{\text{ibr}})$** replaces classical electromechanical swing groups

Coordination algorithm:

$$\mathcal{F}^* = \arg \min_{\mathcal{F}} \sum_{k \in \mathcal{F}} c_k \text{ s.t. fault isolated, load retained}$$

Minimum-cost fault isolation solved via ILP on the updated topology.

WAPC Localisation Results

Metric	Decentralised	WAPC
Fault localisation	71.2%	94.5%
Unnecessary trips	18.4%	4.1%
Load preserved	63.1%	87.6%
Backup response (ms)	420	210

Observation: IBR-aware coherency model reduces unnecessary tripping by -78% relative; backup clearance halved.



Two threat vectors addressed:

- 1 Data integrity attack:** Adversary injects false PMU measurements to manipulate $\hat{k}_{ibr}$ and force mal-operation.

Defence: HMAC-SHA256 on every IED message; relay discards unauthenticated data and holds last trusted $\hat{\theta}$.

- 2 Adversarial ML attack:** Crafted perturbation $\delta\mathbf{x}$ fools PINN classifier ($\epsilon < 5\%$ perturbation causes 40% accuracy drop).

Defence: Adversarial training with FGSM + PGD examples; certified radius $r_{\text{cert}} = 0.04$ (pu).

Cybersecurity Metrics

Metric	No defence	SAMBP-Sec
Attack success rate	67.4%	12.5%
Risk index	1.000	0.451
HMAC overhead (ms)	—	0.3
Adv. PINN accuracy	51.8%	91.4%

Observation: HMAC + adversarial training reduces composite risk index by **−54.9%** with **<0.3 ms** authentication overhead.



Consolidated Performance Across All Modules

Chapter	Module	Key Metric	Baseline	SAMBP
Ch 4	87L adaptive	TPR / FPR	0.783 / 14.8%	1.000 / 0.0%
Ch 5	SyncOC	R^2 / Spurious OC	0.941 / 38	0.999+ / 16 (−58%)
Ch 6	87T SPRT	Inrush rej. / IBR FP	94.1% / 8.7%	99.6% / 0.9%
Ch 7	Distance 21	Correct Z1 trip	74%	96%
Ch 8	Microgrid	Trans. mal-op rate	22%	1.4%
Ch 9	EKF twin	$\hat{k}_{ibr}$ RMSE	—	0.024 pu
Ch 10	PINN	Overall accuracy	81.4%	96.1%
Ch 11	WAPC	Fault localisation	71.2%	94.5%
Ch 12	Security	Risk index reduction	1.000	0.451 (−54.9%)

Observation: All SAMBP modules outperform their fixed-parameter baselines; gains are consistent across $k_{ibr} \in [0, 1.0]$.



Principal contributions:

- 1 **SAMBP framework:** First unified inverse-estimation architecture for all IEC relay types under variable IBR penetration
- 2 **Adaptive 87L:** Perfect TPR=1.000 and zero false trips across full IBR range
- 3 **Non-blocking EKF digital twin:** Real-time parameter oracle with <2-cycle convergence
- 4 **Physics-informed pinn:** +46% accuracy gain on high-IBR faults
- 5 **IBR-aware wapc:** -78% unnecessary trips, $\times 2$ backup speed
- 6 **Hardened cyber layer:** -54.9% composite risk with <0.3 ms overhead

Novelty of the work:

- *Real-time k_{ibr} estimation* — not a planning-level correction
- *Bounded update law* — provably stable under adversarial data injection
- *Stage-2 veto gate* — first formal confidence-gating mechanism for IED trip chains
- Cross-layer design: EKF $\rightarrow$ modules $\rightarrow$ WAPC $\rightarrow$ security — complete protection stack

Future work:

- Hardware-in-loop (RTDS) validation
- Field deployment on TANGEDCO network
- Extension to grid-forming IBR dynamics



Journal papers (submitted / under review):

- ① “Adaptive line-differential protection under variable IBR penetration,” *IEEE Trans. Power Delivery* (under review)
- ② “SAMBP-SyncOC: Self-adaptive over-current coordination for IBR-rich grids,” *IET Generation, Transmission & Distribution* (under review)
- ③ “Physics-informed neural network for fault classification in IBR-dominated networks,” *IEEE Trans. Neural Networks & Learning Systems* (under review)
- ④ “Wide-area protection with IBR coherency: A minimum-cost isolation framework,” *Electric Power Systems Research* (in preparation)

Conference papers (published):

- ① “Non-blocking EKF digital twin for adaptive protection,” *IEEE PES ISGT-Europe 2025*
- ② “Adversarial robustness of PINN-based protection classifiers,” *IEEE TENCON 2025*

Other outcomes:

- 1 patent filed (IIT Madras): “Adaptive relay system with real-time IBR parameter estimation” (2025)
- SAMBP codebase: 4,200+ lines of Python / MATLAB simulation framework
- 2 datasets (fault signatures for IBR grids) uploaded to IEEE DataPort



Thesis Content & Chapter–Publication Mapping

Ch	Title	Pages	Publication / Status
1	Introduction	~18	—
2	Mathematical Foundations	~14	—
3	IBR Characterisation	~16	—
4	Line Differential (87L)	~18	IEEE Trans. PD (under review)
5	Generator OC (SyncOC)	~16	IET GTD (under review)
6	Transformer Diff. (87T)	~14	Conference proceedings
7	Adaptive Distance (21)	~14	—
8	Microgrid Protection	~14	—
9	EKF Digital Twin	~16	IEEE PES ISGT-E 2025
10	ML Fault Classifier (PINN)	~16	IEEE TNNLS (under review)
11	Wide-Area Coordinator	~16	EPSR (in preparation)
12	Cybersecurity	~12	IEEE TENCON 2025
13	Conclusions	~8	—
A–D	Appendices	~12	—
Total (estimated)		~204	



Thank You

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